

Sheet - 9 Functions - 2

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1.Keyword argument

Define a function in Python Introduce , providing their name, age, and profession as arguments.

2.Default argument

WAP to define and utilize a Python function named introduce to print out information about individuals, including their name, age, and profession, with an optional default value for the profession parameter.

3. What is Error in this code and how can we correct this code?

```
def pythagoras(a,b):  
    return square(a) + square(b)  
c2 = pythagoras(3,4)
```

```
def square(x):  
    return x * x
```

```
print("c2 = ", c2)
```

4.What is the output of this code and tell a variable “lang “ is in local or global scope?

```
def local():  
    lang = "Python"  
    print(lang)  
local()
```

5.What is the output of this code and tell a variable “lang “ and “name” is in local or global scope?

```
name = "Suyash"
def test():
    print(name)
    lang = "Python"
    print(lang)
test()
print(name)
```

6.What is Error in this code and how can we correct this code?

```
def test():
    lang = "Python"
    print(lang)
test()
print(lang)
```

7.What is Error in this code and how can we correct this code?

```
name = "Suyash"
print(name)
def fun():
    print(name)
    name = "Chaudhary"
    lang = "Python"
    print(lang)
fun()
print(name)
```

8.What is the output of this code?

```
name = "Suyash"
print(name)
def test():
    global name
    print(name)
    name = "Chaudhary"
    print(name)
test()
print(name)
```

9.What is the output of this code?

```
name = "abc"
print(name)

def fun1():
    name = "def"
    print(name)

print(name)
fun1()

def fun2():
    global name
    name = "ghi"
    print(name)

print(name)
fun2()
print(name)
```

10.What is the output of this code?

```
def fun1():  
    name = "Suyash"  
    def fun2():  
        name = "Chaudhary"  
    fun2()  
    print(name)  
fun1()
```

11.What is the output of this code?

```
def fun1():  
    name = "Suyash"  
    def fun2():  
        nonlocal name  
        name = "Chaudhary"  
    fun2()  
    print(name)  
fun1()
```

12.Square of that integer using a lambda function

You want to create a Python program that takes an integer input, calculates the square of that integer using a lambda function, and then prints the result.

Input :
a= 4

Output:
16

13.Odd and Even No

Write a Python program that performs the following tasks:

Accepts an integer input from the user and stores it in a variable num.

Defines a lambda function even_odd_No that takes one parameter num and returns "even" if num is even, otherwise it returns "odd".

Calls the lambda function even_odd_No with the input num and prints the result.

Provide the Python code to implement the above requirements.

14.Max of two No.

Write a Python program that defines a lambda function called Max that takes two parameters a and b, and returns the maximum of the two values. Then, call the lambda function Max with the values 1 and 2, and print the result obtained from the call in a single line of code.

15.Max of three No.

Create a Python program that accomplishes the following tasks:

Define a lambda function max_of_three that takes three parameters a, b, and c, and returns the maximum of the three values.

Call the lambda function max_of_three with three integer values.

Print the result obtained from calling the lambda function.

NOTE:- Try to Solve this question with if-else.

16.Filter Even No

Write a Python program that defines a function fun(a) which takes an integer a as input and returns True if a is even, otherwise it returns False. Then, create a list called sequence containing integers and floating-point numbers. Use the filter() function along with the fun function to filter out the even numbers from the sequence list, and store the filtered values in a new list called filtered. Finally, print the filtered list.

17.Filter Even No with lambda fuction

Create a Python program that accomplishes the following tasks:

Define a list arr containing integer values.

Use the filter() function with a lambda function to filter out the even numbers from the list arr.

Store the filtered values in a variable ans.

Print the ans list.

18.Filter Numbers Greater Than 60

Write a single line of Python code that accepts a list of integer values from the user, filters out the numbers greater than 60 using the filter() function with a lambda function, and stores the filtered values in a variable ans. Print the ans list.

18.Add 10 to Each Element of a List Using Map and Lambda Function

Create a Python program that accomplishes the following tasks:

Define a list arr containing integer values.

Use the map() function with a lambda function to add 10 to each element of the list arr.

Store the modified values in a variable ans.

Print the ans list.

19.Convert Names to Lowercase Using Map and Lambda Function

Create a Python program that accomplishes the following tasks:

Define a list names containing string values.

Use the map() function with a lambda function to convert each name in the list names to lowercase.

Store the modified names in a variable result.

Print the result list.

20. Python Program Using map, filter, and lambda Functions

Write a Python program that utilizes the map, filter, and lambda functions to achieve the following objectives:

Use map to convert a list of integers to their corresponding squares.

Use filter to keep only the squared numbers that are multiples of a specific number.